

Questions and Answers with
Distinguished Guest Dr. Jeffrey N. Heinz:
The Role of Mathematical Linguistics, the Rise of A.I.,
and His Favorite Projects

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Dr. Jeffrey N. Heinz is a Professor at Stony Brook University with a joint appointment in the Department of Linguistics and the Institute of Advanced Computational Science. He conducts research in several related areas including theoretical, computational and mathematical linguistics, grammatical inference, computational learning theory, theoretical computer science, robotic planning and control, and artificial intelligence. His research has been published in the Journal of Language Modelling, Linguistic Inquiry, Theoretical Computer Science, the Transactions of the ACL, and Science. He has co-authored a book on grammatical inference for computational linguists, co-edited three books, and co-guest-edited special issues of the journals Machine Learning and Phonology.

Professor Heinz graduated in 1996 from University of Maryland with a B.S. in Mathematics and a B.A. in Linguistics. He obtained his Ph.D. in Linguistics from UCLA in 2007 and spent ten years as a professor at the University of Delaware before coming to Stony Brook in 2017. The Linguistic Society of America recognized Professor Heinz with its 2017 Early Career Award for his “Contributions leading to a new computational science of inference and learning as applied to language.” In 2022, he received SUNY Chancellor’s Award for Excellence in Scholarship and Creative Activities. When not busy teaching and researching, Professor Heinz enjoys swimming, playing chess, solving puzzles, and programming.

Professor Heinz’s answers have been taken from an interview conducted by Shaon Anwar and have been edited for length and clarity.

1) Linguistics is an inherently interdisciplinary field. With its many subfields and connecting disciplines to focus on, you are investigating the sound structure of language from a mathematical and computational perspective. What drew you to this area of research?

When I first took a linguistics course as an undergraduate, I felt phonology was, in some ways, a bit more concrete than other areas like syntax. It was partly because I wanted to be a mathematician, so I came into linguistics for mathematics. I was kind of skeptical about what professors were saying about syntax and tree structures, whereas in phonology, it just seemed a lot more clear.

The other part of it was that in phonology, a lot of it is completely unconscious. When I'm talking, my stress patterns are a certain way, everything is a certain way, but no one taught me that. I read a lot of Bertrand Russell when I was your age. He wondered how it is that humans, whose encounters with the world are so brief and so fragile, come to know as much as they know. That was really striking to me. One of the things we know is our language, which is actually quite an intricate set of knowledge that we have, and again, a lot of it is unconscious. So it seemed to me that trying to understand how we come to acquire that phonological knowledge was a concrete way we could begin to ask some parts of the question that Russell was asking.

2) Did studying mathematics as an undergraduate student prepare you in the field of linguistics?

I think it helped me prepare a lot! I came in as an undergraduate only wanting to study math. I wanted to be a pure mathematician, and like I mentioned before, I was interested in Russell, so I was interested in Proof Theory. I felt like at the time, I hit a wall in mathematics. I was in an advanced class, and I didn't do well in it. I withdrew from the class, and I felt like my dream of being a mathematician was over. Then, I took a linguistics course, and I thought it was an easy course, but I could also see that there was a system there. Mathematics is really just a way of thinking about things. Obviously there's knowledge there, but it's really a kind of attitude more than anything: how you approach something.

I will also add that I had the pleasure of working with a lot of students who have done a lot of work in mathematics and went into linguistics. I find that, if they're interested in languages, the linguistics courses offer no resistance. They can understand it so quickly. Linguistics, when compared to mathematics or physics, is a relatively young field, so you can really get to the cutting edge with just a couple of years of study, and not like in physics or mathematics where you really have to be an advanced student in high school, and advanced student in college, get a Ph.D., and do several post-docs.

3) How would you describe mathematical linguistics?

Linguistics is often described as the scientific study of language, and so we should think of language really as a natural phenomenon. In some ways it's a biological phenomenon because we are biological creatures, and we have the capacity to communicate at a very high level. We're talking about abstract ideas where even the notion of a word is an abstract concept. So, there's something interesting going on here about languages.

Language is also systematic; it's not just the case that any sequence of words is going to form

a sentence in English. So what is that system? How do we explain why some sentences can be interpreted one way, but not some other way? These are the kinds of questions that linguistics are interested in. What is the underlying system that governs language? Mathematical linguistics brings the mathematical way of thinking to these questions.

For example, we want to build a translator that expresses sentences in one language into another language. Ultimately, a language is a sequence of words or symbols. Think about when you study mathematical functions. First, you learn about linear functions, polynomial functions, or quadratic functions. Then you learn about trigonometric functions. By the time you get to calculus and integrations, you have a very rich vocabulary to talk about numerical functions. Now, consider functions that take strings, sequences of symbols, and convert them into other sequences of symbols, which is fundamentally what translation is like. What kind of function is that? It's not a numerical function; it's a function over sequences. Unlike the study of numerical functions, the study of functions of sequences is less than one hundred years old. Now, I would say we have little understanding of these functions.

It's almost like saying, "Well, we've got linear functions and nonlinear functions." It's a bit more than that, but it's an approximation. We're actually getting a much more fine-grain understanding of the types of string-to-string functions that there are. When we have that, we can say things like, "When we're translating from this language to this language, it's this kind of function," which is important! How does gravity work? Well, it's an inverse square law. So we need to know what that means. Right now, we're coming up with the vocabulary for the mathematics behind words that would mean things kind of like "inverse square law," but for functions over sequences instead of functions over numbers. That's one example of what mathematical linguistics is about.

4) Earlier you mentioned translation software. Recently, artificial intelligence breakthroughs including the AlphaFold software and ChatGPT have made waves in the scientific world. How will this flurry of innovations create solutions to problems in linguistics?

We have to understand what a solution really is. What ChatGPT does is obviously very impressive, but we can ask, "What problem is it solving?" And that's a harder question to answer. Again, ChatGPT has been trained on incredible amounts of data. That's not what human children are doing when they learn. You weren't exposed to every book that's in the Google Book's library that's written in English. You learned to speak and generate and understand sentences that you never heard before within the first few years of your life. Your linguistics experience was sufficient for you to get a fairly good command of the English grammar. That's not how ChatGPT works. ChatGPT is doing a lot of statistical inference, and I'm not saying statistics is the wrong answer; I think statistics is going to be part of the answer. The problem with statistical methods is that rare events are difficult to deal with, and when you're talking about the kinds of sentences that we can understand and produce, there are a lot of rare events.

For example, a Zipfian Curve is an inverse power law, so the frequent things are very common, but the infrequent things actually dominate. A common example has to do with the frequency of words in texts. If you take any English text and count the number of times each word occurs in the text, you'll find the word "the" is by far the most common English word. Then, if you go down to the second word "of," it occurs half as much. The point is that as you keep doing this, you'll find that most words in the text occur exactly once. Most constructions occur just once, and so

how do you distinguish between the things that are rare and just occur once, and the things that are impossible? What these systems are good at is the first part of the inverse power law; they're not good at the tail.

Now, have they made progress? Yes, I would say there's definitely been progress, but the idea that somehow there's nothing else to do is not true at all. The other side is: are they fundamentally doing the right kind of work that's going to lead us to a genuine solution? That, I think the jury is out on. I think it's easy to say that they have to take this further than anything else, but look at the teams that are behind these teams. If I had those same resources that those teams have, using other kinds of methods, I think I can get pretty far too! We can't compete with them; they've got incredible resources behind them, and they're pursuing a particular line of research. They're getting somewhere - that's great. I just think that there's another way to go that will help us get real solutions to some of these problems. It's fine to disagree.

5) What are some of your favorite projects?

I loved working on my dissertation because I was answering a question about phonological patterns and how they could be learned. Since then, I've worked on projects about kinds of string-to-string functions; something that will have a very lasting impact. We're in the process of figuring out the fundamentals of it now. Again, one of the exciting things about mathematics is that you get to figure this stuff out, and if you get it right, it's there. Ken Dill, who is one of the faculty members at Stony Brook University, had come up with a very simple way of posing a protein folding project that I worked on, which is a very simple puzzle. You consider a sequence of red and blue dots - beads on a string. That's your sequence of amino acids. Basically your amino acids come in two basic types, the polar ones and the hydrophobic ones. One of the main factors in protein folding is that the hydrophobic amino acids - the blue ones on your string - want to clump together. So now you have a sequence of blue and red dots and you want to fold them up to maximize the number of blue-on-blue contacts. His proposal was to put them in a 2D grid space, and now you want to put this string somehow along the grid so that the number of blue dots that are adjacent is maximized. That's actually a really hard problem, but you can use computational linguistics methods for parsing. What do we do in computational linguistics? We have a sequence, and we try to get a structure out of it because the structure tells us something about how we interpret it and how we understand it. That's what we're doing in this protein folding problem; we have a sequence of amino acids and we want to get a structure out of it: we want to fold it up into a 3D object. The structures are different in linguistics and protein folding, but the fundamental problem of taking a sequence and getting a structure out of it is very, very similar. They use the same kind of methods, these parsing methods, and if you can find a way to efficiently parse these amino acids into structures, it's going to be huge.

One of the great things about working on mathematical linguistics is that mathematics, because you've abstracted a way to a certain degree, allows me to work on these interdisciplinary topics, and it allows me to communicate with the protein folding guys, and the robotics and planning guys. It's been fascinating to me. I've learned so much, and I love it! It's one of the great things about the institute I'm in at Stony Brook; it's an interdisciplinary institute, and that's been really fabulous.

6) What are the challenges you encounter in your field of research?

I mentioned before that in linguistics there are different communities of people who have different opinions about things. Frankly, a lot of people don't think the mathematical aspect is very useful. For them, it's too abstract. They think it doesn't connect with reality. What I would say to them is that everything is abstract! A lot of them would prefer to use statistics or real numbers or things like that. But, numbers are abstract. Real numbers are especially abstract. The idea that somehow what we're doing is too abstract, I don't really understand.

I will say that within and beyond the linguistics field, people want to see applied research. There's nothing wrong with applied research - doing applied research is great, a lot of technological advancement we have in the last 50 years is coming from applied research, but I don't think applied research is the only kind of research that should be done. I think theoretical research should also be done, where the horizon is a bit further down the road. The work that I'm doing now, these fine-grained understandings of functions over sequences is going to be integrated into all the technology not in five years, not in ten years, it's going to be like 30 years from now. It's going to take a while for it to get established. It's going to take a while for people to realize it's out there, then they will start using it, then it will get integrated. So, some of the challenges I face in my own research is that people are skeptical of the mathematical stuff. My own view is that there are many ways to search for the truth, and people should welcome different approaches even if they are more theoretical than what they are comfortable with.

7) Lastly, what is your advice to Stuyvesant students who would like to pursue computational and mathematical linguistics?

You're so lucky! It's a great time to do that. The mathematical and computational stuff is more accessible than it has ever been before. Keep an open mind, study the questions that interest you, learn as much math and computer science as you can, but also learn as much about languages as much as you can! I had a professor, Ed Keenan, who was a mathematical linguist at UCLA, and he would say, "You should read grammar like you would read science fiction." You should go into the library; there are stacks and stacks of shelves and there are whole shelves that are filled with grammar. You start looking at it and ask yourself: "What does this language do? How does it do this thing? What does it do here?" It's almost like a botanist studying the different kinds of things plants do. You want to get a sense of that, because then all the mathematical and computational concepts will begin to sort through them in your mind. I would say that it's a great field to go into.